

ALS Study Companion

Philippine Alternative Learning System
Complete Structured Plan

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1. Project Overview & Current State Analysis

The ALS Study Companion is an offline-first, multi-platform Learning Management System (LMS) built for the Philippine government's Alternative Learning System program. It currently uses Flutter/Dart with Supabase as the backend, targeting both mobile (Android/iOS) and web platforms. Below is a high-level assessment of the current codebase before the restructuring plan is applied.

1.1 What Is Correctly Built

- Monorepo structure using Flutter workspaces (ALS-LMS root with apps/ and packages/).
- Supabase backend integration: Auth, PostgreSQL with RLS, and Storage buckets.
- Offline-first architecture using sqflite and Drift ORM with a SyncQueue mechanism.
- Shared packages: shared_models, shared_services, shared_ui for code reuse.
- Database triggers: handle_new_auth_user, update_updated_at_column, validate_student_lrn.
- Role-based access: Student, Teacher, and Admin roles with separate dashboards.
- Biometric login via local_auth and secure storage via flutter_secure_storage.
- Row Level Security (RLS) policies across all Supabase tables.
- Conflict resolution via last-write-wins on updated_at during sync.

1.2 What Needs to Be Fixed, Added, or Renamed

- The mobile app folder is named student_phone but serves BOTH students and teachers. Must be renamed to mobile_app.
- The ALS Center registration flow (school/center self-registration) is completely missing. The notes describe a public-facing registration page for centers, but no such page, database table column, or approval workflow exists.
- The course/subject enrollment system is incomplete. Students should enroll via QR code or PIN, but there are no course tables, QR code generation, or PIN assignment in the schema.
- Teachers cannot currently structure courses with timelines. There is no timeline or schedule model tied to a course.
- Core Academic Subjects are not being registered per-center. Centers need to declare which subjects they offer.
- Virtual certificate generation on course completion is absent from the current implementation.
- The admin_web app for System Admins is missing the ALS Center registration review/approval workflow. It only has a "Create Center" form, not a review queue.
- shared_models and shared_services package names differ from the README's shared_core and backend_services naming. These must be reconciled.
- Two separate SQLite databases (als_study_companion.db via sqflite AND als_local.sqlite via Drift) create redundancy. A unified approach is needed.
- Teacher registration does not associate the teacher to an ALS center at registration time. This must be added.

- Student registration does not associate the student to an ALS center at registration time. This must be added.

2. Terminology, Naming Conventions & Folder Renames

Consistent naming is critical for code readability, team collaboration, and correct routing. The following table defines the authoritative terms to use everywhere: in folder names, class names, variable names, database columns, UI labels, and comments.

2.1 Folder & Package Renames

Current Name	Correct Name	Reason
apps/student_phone	apps/mobile_app	The app is used by both Students AND Teachers. "student_phone" is misleading and will cause confusion.
packages/shared_models	packages/shared_core	Aligns with README documentation and better communicates it holds models, enums, constants, and utils.
packages/shared_services	packages/backend_services	Clearly signals it wraps all Supabase/backend interactions.
packages/shared_ui	packages/shared_ui	No change needed. Name is accurate.
apps/admin_web	apps/admin_web	No change needed. Name is accurate.

2.2 Terminology Standards

Use these exact terms consistently across the entire codebase, UI labels, database columns, and documentation:

Use This Term	Do Not Use	Context
ALS Center	School, Branch, Location, Center	The registered institution within the ALS program.
Center Admin	School Admin, ALS Admin, Branch Admin	The admin user belonging to a specific ALS Center.
System Admin	Developer, Super Admin, Admin (generic)	The Anthropic-side admin who reviews center registrations.
Course	Lesson Plan, Class, Module Set	A teacher-created grouping of lessons with a timeline. Distinct from a single Lesson.
Lesson	Module, Material, File	A single learning unit within a Course (has video/PDF/study guide).
Subject	Core Subject, Category, Track	A core academic subject offered by an ALS Center (e.g., Mathematics, Filipino).
Enrollment	Joining, Adding, Registration (for course)	The act of a student joining a Course via QR or PIN.

Use This Term	Do Not Use	Context
Course PIN	Code, Key, Password	A unique short alphanumeric code for a Course used for enrollment.
Teacher Approval	Verification, Activation	Admin approval step after teacher registers.
Center Registration	Center Signup, Center Creation	The public-facing form where an ALS Center applies to join.
Learning Material	File, Resource, Attachment	Any downloadable file attached to a Lesson (PDF, video, doc).
Virtual Certificate	Badge, Diploma, Reward	The in-app certificate shown on course completion.
Learner Reference Number (LRN)	Student ID, DepEd Number	The official 12-digit student identifier used in Philippine ALS.

3. Corrected System Architecture

3.1 Monorepo Folder Structure (Final)

The following is the complete, authoritative folder structure after all renames and additions:

```

ALS-Study-Companion/  ALS-LMS/                                <- Monorepo root
apps/                admin_web/                               <- Flutter Web: System Admin +
Center Admin         mobile_app/                               <- Flutter Mobile: Students
& Teachers [RENAMED] packages/                                shared_core/          <-
Pure Dart models, enums, constants [RENAMED]                 backend_services/     <-
<- Supabase wrappers & sync [RENAMED]                         shared_ui/            <-
Design system & reusable widgets supabase/                   config.toml           migrations/
<- All SQL migrations in order docs/                          CoreAcademicSubjects.pdf
SETUP.md             ARCHITECTURE.md
  
```

3.2 Dependency Graph

The dependency graph below shows what each package/app imports. No circular dependencies are allowed.

```

shared_core  (zero internal deps)      ^      | backend_services  (depends
on: shared_core)      ^      |-----|      admin_web
mobile_app  (shared_core +      (shared_core +  backend_services +
backend_services +  shared_ui)      shared_ui)
  
```

4. User Roles, Registration Flows & Permissions

4.1 All User Roles (Expanded)

The system now has four distinct roles, not three. The Center Admin is a new role split from the generic "Admin".

Role	Platform	Description
System Admin	admin_web (web)	Us (the developers). Reviews and approves ALS Center registration requests. Has full system access.
Center Admin	admin_web (web)	Admin of a specific, already-approved ALS Center. Manages their center's subjects, teachers, and students.
Teacher	mobile_app (mobile)	Registered and approved teacher within an ALS Center. Creates courses and uploads learning materials.
Student	mobile_app (mobile)	Registered student at an ALS Center. Enrolls in courses via QR/PIN and views learning materials.

4.2 ALS Center Registration Flow (NEW)

NEW FLOW — This entire flow does not yet exist and must be built from scratch.

1. A representative of an ALS Center visits the web registration page (admin_web public route: /register-center).
2. They fill out the Center Registration Form: Center Name, Address, Region, Contact Number, and Admin Account details (full name, email, password).
3. On submit, a new record is created in the als_center_registrations table with status = "pending".
4. The System Admin logs into admin_web and sees a "Pending Registrations" queue on their dashboard.
5. System Admin reviews the request. They can Approve or Reject.
6. On Approval: the als_centers record is activated (is_active = true), the submitted admin account is created in Supabase Auth with role = "center_admin", and a confirmation email is sent to the center.
7. On Rejection: the system sends a rejection email with a reason and the record is marked as rejected.
8. Once approved, the ALS Center appears in the dropdown on the mobile_app registration screen under "Select your ALS Center".

4.3 Teacher Registration Flow (Updated)

9. Teacher opens mobile_app, selects the "Teacher" toggle on the registration screen.
10. Teacher selects their ALS Center from the dropdown (only approved centers show here).
11. Teacher fills in: Full Name, Email, Password, Phone Number.
12. Teacher taps "Create Account". Account is created with teacher_verified = false and assigned to the chosen als_center_id.
13. Teacher is shown an "Awaiting Approval" screen. The Center Admin of that specific ALS Center is notified.
14. Center Admin (not System Admin) logs into admin_web and approves the teacher from their center's Teacher Management panel.
15. On approval: teacher_verified is set to true. Teacher receives a push notification or email.
16. Teacher can now log in and access the Teacher Dashboard.

4.4 Student Registration Flow (Updated)

17. Student opens mobile_app, selects "Student" toggle on the registration screen.
18. Student selects their ALS Center from the dropdown.
19. Student fills in demographic form: Full Name, Email, Password, LRN (12 digits), Date of Birth, Phone, Occupation, Last School Attended, Last Year Attended.
20. Student taps "Create Account". No approval needed. Student is immediately active.
21. Student is navigated to their dashboard and prompted to enroll in a course.

4.5 Course Enrollment Flow (NEW)

NEW FLOW — This entire flow does not yet exist and must be built from scratch.

22. After login, the student's dashboard shows an "Enroll in a Course" prompt if they have no active courses.
23. The student taps "Enroll" and is presented two options: Scan QR Code or Enter Course PIN.
24. Option A — QR Scan: The app opens the device camera. The teacher's QR code encodes the Course PIN. On scan, the student is enrolled.
25. Option B — Enter PIN: The student manually types the 6-character alphanumeric Course PIN.
26. On successful enrollment: a new record is added to the course_enrollments table linking the student to the course.
27. The student now sees all lessons in that course, the course timeline, and their completion progress.

5. Database Schema — Additions & Corrections

5.1 New Tables Required

The following tables must be added to the Supabase (PostgreSQL) schema. SQL migration files should be created for each.

Table: als_center_registrations

PURPOSE	Stores ALS Center registration requests awaiting System Admin review.
Column	Definition
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()
center_name	TEXT NOT NULL
address	TEXT NOT NULL
region	TEXT NOT NULL
contact_number	TEXT NOT NULL
admin_full_name	TEXT NOT NULL -- The person submitting the request
admin_email	TEXT NOT NULL UNIQUE
status	TEXT NOT NULL DEFAULT 'pending' CHECK (IN ('pending', 'approved', 'rejected'))
rejection_reason	TEXT -- Filled in if rejected
reviewed_by	UUID REFERENCES users(id) ON DELETE SET NULL -- System Admin who reviewed
reviewed_at	TIMESTAMPTZ
created_at	TIMESTAMPTZ DEFAULT NOW()

Table: courses

PURPOSE	A teacher-created course grouping lessons under a core subject, with a defined timeline and enrollment mechanism.
Column	Definition
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()
title	TEXT NOT NULL
description	TEXT

Column	Definition
subject_id	UUID NOT NULL REFERENCES center_subjects(id) ON DELETE CASCADE
teacher_id	UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
als_center_id	UUID NOT NULL REFERENCES als_centers(id) ON DELETE CASCADE
course_pin	TEXT NOT NULL UNIQUE -- 6-character alphanumeric, generated on creation
qr_code_url	TEXT -- Storage URL of the generated QR code image for this course
start_date	DATE
end_date	DATE
is_active	BOOLEAN DEFAULT TRUE
is_published	BOOLEAN DEFAULT FALSE -- Must be true for students to enroll
sync_status	TEXT DEFAULT 'synced'
created_at	TIMESTAMPTZ DEFAULT NOW()
updated_at	TIMESTAMPTZ DEFAULT NOW()

Table: course_enrollments**PURPOSE**

Links a student to a course. Created when a student scans the QR or enters the PIN.

Column	Definition
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()
course_id	UUID NOT NULL REFERENCES courses(id) ON DELETE CASCADE
student_id	UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
enrolled_at	TIMESTAMPTZ DEFAULT NOW()
completed_at	TIMESTAMPTZ -- Set when all lessons are 100% complete
is_active	BOOLEAN DEFAULT TRUE
	UNIQUE(course_id, student_id) -- Prevent duplicate enrollments

Table: center_subjects**PURPOSE**

Stores the Core Academic Subjects that a specific ALS Center has registered to offer (from the CoreAcademicSubjects.pdf list).

Column	Definition
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()
als_center_id	UUID NOT NULL REFERENCES als_centers(id) ON DELETE CASCADE
subject_name	TEXT NOT NULL -- e.g., "Mathematics", "Filipino", "English"
subject_code	TEXT NOT NULL -- Short code e.g., "MATH", "FIL"
grade_level	TEXT -- e.g., "Elementary", "Junior High School"
is_active	BOOLEAN DEFAULT TRUE
created_at	TIMESTAMPTZ DEFAULT NOW()
	UNIQUE(als_center_id, subject_code, grade_level)

Table: course_timeline**PURPOSE**

Stores the timeline items for a course (weeks, milestones, lesson order).
Displayed to students.

Column	Definition
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()
course_id	UUID NOT NULL REFERENCES courses(id) ON DELETE CASCADE
title	TEXT NOT NULL -- e.g., "Week 1: Introduction to Fractions"
description	TEXT
lesson_id	UUID REFERENCES lessons(id) ON DELETE SET NULL -- Optional: links to a lesson
start_date	DATE
end_date	DATE
order_index	INT NOT NULL DEFAULT 0
created_at	TIMESTAMPTZ DEFAULT NOW()

Table: certificates**PURPOSE**

Generated when a student completes 100% of a course. Stores the certificate data for rendering.

Column	Definition
id	UUID PRIMARY KEY DEFAULT gen_random_uuid()

Column	Definition
enrollment_id	UUID NOT NULL UNIQUE REFERENCES course_enrollments(id) ON DELETE CASCADE
student_id	UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
course_id	UUID NOT NULL REFERENCES courses(id) ON DELETE CASCADE
issued_at	TIMESTAMPTZ DEFAULT NOW()
certificate_url	TEXT -- Optional: storage URL if exported to PDF

5.2 Changes to Existing Tables

Table: als_centers — Add Column

Column to Add	Definition & Reason
registration_id	UUID REFERENCES als_center_registrations(id) -- Links back to the registration request that created this center.
center_admin_id	UUID REFERENCES users(id) -- The Center Admin user for this specific center. Foreign key to users table.

Table: users — Add Column

Column to Add	Definition & Reason
role (update CHECK)	Change CHECK constraint to: CHECK (role IN ('student', 'teacher', 'center_admin', 'system_admin')) -- Add new roles.

Table: lessons — Add Column

Column to Add	Definition & Reason
course_id	UUID REFERENCES courses(id) ON DELETE CASCADE -- Every lesson must belong to a Course.

6. Code Changes — shared_core Package

6.1 New Models to Add

Add the following model files under packages/shared_core/lib/models/:

CourseModel

- File: course_model.dart
- Fields: id, title, description, subjectId, teacherId, alsCenterId, coursePin, qrCodeUrl, startDate, endDate, isActive, isPublished, syncStatus, createdAt, updatedAt.
- Methods: fromMap(), toMap(), copyWith(). The fromMap() must handle both snake_case (Supabase) and camelCase (SQLite).

CourseEnrollmentModel

- File: course_enrollment_model.dart
- Fields: id, courseId, studentId, enrolledAt, completedAt, isActive.

CenterSubjectModel

- File: center_subject_model.dart
- Fields: id, alsCenterId, subjectName, subjectCode, gradeLevel, isActive, createdAt.

CourseTimelineModel

- File: course_timeline_model.dart
- Fields: id, courseId, title, description, lessonId (nullable), startDate, endDate, orderIndex, createdAt.

CertificateModel

- File: certificate_model.dart
- Fields: id, enrollmentId, studentId, courseId, issuedAt, certificateUrl (nullable).

AlsCenterRegistrationModel

- File: als_center_registration_model.dart
- Fields: id, centerName, address, region, contactNumber, adminFullName, adminEmail, status (CenterRegistrationStatus enum), rejectionReason, reviewedBy, reviewedAt, createdAt.

6.2 Updated Enums

UserRole — Update

Add two new values to the existing UserRole enum:

- centerAdmin — for ALS Center admins.
- systemAdmin — replaces the generic "admin" role for the developers/us.

Keep the existing admin value temporarily during migration, but deprecate it.

CenterRegistrationStatus — New

- pending — default, awaiting review.
- approved — System Admin approved it.
- rejected — System Admin rejected it.

6.3 Updated Constants (DbConstants)

Add the following table name constants to DbConstants:

- tableCourses = 'courses'
- tableCourseEnrollments = 'course_enrollments'
- tableCenterSubjects = 'center_subjects'
- tableCourseTimeline = 'course_timeline'
- tableCertificates = 'certificates'
- tableAlsCenterRegistrations = 'als_center_registrations'

7. Code Changes — backend_services Package

7.1 New Service Methods

In SupabaseDatabaseService — Add:

- `getCoursesByCenter(alsCenterId)` — Fetches all published courses for an ALS Center.
- `getCoursesByTeacher(teacherId)` — Fetches a teacher's own courses.
- `saveCourse(course)` — Upserts a `CourseModel`.
- `deleteCourse(courseId)` — Deletes a course and cascades to enrollments.
- `enrollStudentInCourse(courseId, studentId)` — Creates a `CourseEnrollmentModel`.
- `getEnrollmentsByStudent(studentId)` — Fetches all enrollments for a student.
- `getCenterSubjects(alsCenterId)` — Fetches all subjects a center offers.
- `saveCenterSubject(subject)` — Upserts a `CenterSubjectModel`.
- `getCourseTimeline(courseId)` — Fetches all timeline items for a course.
- `saveCourseTimelineItem(item)` — Upserts a `CourseTimelineModel`.
- `issueCertificate(enrollmentId, studentId, courseId)` — Creates a `CertificateModel`.
- `getCenterRegistrations({status})` — For System Admin to fetch registration requests by status.
- `approveCenter(registrationId)` — Updates registration status to approved and activates the `als_centers` row.
- `rejectCenter(registrationId, reason)` — Updates registration status to rejected with a reason.
- `generateCoursePIN()` — Generates a unique 6-character alphanumeric code and verifies uniqueness against the `courses` table.

In SupabaseStorageService — Add:

- `uploadQrCode(courseId, qrImageBytes)` — Uploads QR code PNG to storage bucket `qr-codes/courses/{courseId}/qr.png`.

7.2 New Storage Bucket

**NEW
BUCKET**

`qr-codes` — Stores QR code images for course enrollment. Path: `courses/{courseId}/qr.png`. Public read access. Write restricted to authenticated teachers.

8. Code Changes — mobile_app (Formerly student_phone)

8.1 Registration Screen — Updates

- Add ALS Center dropdown to both Student and Teacher registration forms. This dropdown is populated from getAlsCenters() filtering is_active = true only.
- Store the selected als_center_id in the user's metadata when calling Supabase Auth signUp. This value must be included in user_metadata so the handle_new_auth_user DB trigger can store it in public.users.
- Teacher registration: add an "I am a teacher at [center dropdown]" label for clarity.

8.2 New Teacher Features

Course Creation (New Screen: TeacherCourseCreateView)

28. Teacher selects a Core Academic Subject from the center_subjects list (subjects their center offers).
29. Teacher enters: Course Title, Description, Start Date, End Date.
30. On save, a CourseModel is created with a generated coursePin and a QR code is generated and uploaded.
31. Teacher can view the QR code and the Course PIN from the Course Details screen.
32. Teacher can share the PIN/QR to students verbally, via messaging, or by displaying it.

Course Timeline Builder (New Screen: TeacherCourseTimelineView)

33. Teacher can add, reorder, and delete timeline items within a course.
34. Each timeline item has: Title, Description, optional linked Lesson, Start Date, End Date.
35. The timeline is displayed chronologically to students.

Updated Teacher Dashboard Tab Structure

Tab	Content
Home	Welcome, quick stats, quick-action grid (New Course, Add Lesson, Create Quiz, Schedule, Announce).
Courses	List of teacher's courses. Tap a course to see its lessons, timeline, enrolled students, and QR/PIN.
Students	List of enrolled students across all courses. View individual progress.
Sessions	Schedule and manage teaching sessions.
Announce	Broadcast announcements to enrolled students.

8.3 New Student Features

Course Enrollment Screen (New: StudentEnrollView)

- 36. Accessible from student home screen via "Enroll in a Course" button.
- 37. Two tabs: Scan QR and Enter PIN.
- 38. Scan QR tab opens the device camera using `mobile_scanner` or `qr_code_scanner` package.
- 39. Enter PIN tab shows a 6-character text input. Validates that the PIN exists and the course is published and active.
- 40. On success, student is enrolled and navigated to the course view.

Course View (New: StudentCourseView)

- 41. Shows Course Title, Subject, Teacher Name, and Center.
- 42. Shows the course timeline with each item and its date range.
- 43. Shows all lessons within the course, each with a completion indicator.
- 44. Shows overall course completion progress as a live percentage (from `ProgressModel` records).
- 45. Shows a "Complete" badge and the Virtual Certificate button when all lessons are 100%.

Virtual Certificate (New: StudentCertificateView)

- 46. Triggered automatically when course completion reaches 100%.
- 47. Displays a styled in-app certificate with: Student Full Name, Course Title, Subject, ALS Center Name, Completion Date, Teacher Name.
- 48. The certificate is also stored in the certificates Supabase table via `issueCertificate()`.
- 49. Student can take a screenshot or download as PDF (use Flutter's screenshot or pdf package).
- 50. A banner notification is shown on the dashboard ("You completed [Course]! View your certificate.").

Updated Student Dashboard Tab Structure

Tab	Content
Home	Welcome message, enrolled courses summary, completion banners, and "Enroll in a Course" button.
Courses	List of enrolled courses with progress indicators. Tap to open <code>StudentCourseView</code> .
Materials	Aggregated view of all lesson materials across enrolled courses.
Downloads	Downloaded materials available offline.

8.4 SQLite Schema Additions (Local Database)

Add the following tables to `DatabaseHelper` (sqlite) to support offline-first functionality for the new features:

- `courses` — Mirror of the Supabase `courses` table with `syncStatus` column.
- `course_enrollments` — Mirror of the Supabase `course_enrollments` table.
- `center_subjects` — Mirror of `center_subjects`, used to populate the subject dropdown offline.
- `course_timeline` — Mirror of `course_timeline` for offline course view.
- `certificates` — Mirror of `certificates` for offline certificate viewing.

8.5 New Dependencies to Add (pubspec.yaml)

Package	Reason
<code>mobile_scanner: ^3.5.0</code> OR <code>qr_code_scanner: ^1.0.1</code>	QR code scanning for course enrollment.
<code>qr_flutter: ^4.1.0</code>	QR code display/generation on teacher side.
<code>screenshot: ^2.1.0</code>	Certificate screenshot/export feature.
<code>pdf: ^3.10.8</code> + <code>printing: ^5.11.0</code>	Certificate PDF generation and printing.

9. Code Changes — admin_web

9.1 Two Admin Roles in admin_web

The admin_web app must now support two distinct admin roles: System Admin and Center Admin. On login, the app detects the user's role and shows the appropriate navigation/dashboard.

Feature	System Admin Sees	Center Admin Sees
Dashboard	Global stats: all centers, all teachers, all students.	Center-specific stats: their teachers, students, courses.
Center Registrations	Pending Registration queue with Approve/Reject.	Not visible.
Center Management	Full CRUD on all centers.	View/edit their own center info only.
User Management	All users across all centers.	Only users within their center.
Subject Management	Read only.	Can add/remove Core Academic Subjects their center offers.
Teacher Approval	Can approve teachers at any center.	Can approve teachers at their own center only.
Content Moderation	Can delete/unpublish any content.	Can manage content within their center.
Audit Logs	Full audit log.	Filtered to their center's actions only.

9.2 New Pages & Features

CenterRegistrationReviewPage (New — System Admin Only)

- Accessible from System Admin sidebar: "Registrations" nav item.
- Shows a list of all als_center_registrations with status = "pending".
- Each card shows: Center Name, Region, Admin Name, Admin Email, Submitted Date.
- Approve button: calls approveCenter(). This sets status = "approved", is_active = true in als_centers, and creates the Center Admin user account.
- Reject button: opens a dialog to enter a rejection reason. Calls rejectCenter(registrationId, reason).

SubjectManagementPage (New — Center Admin)

- Center Admin can see all Core Academic Subjects from the master list (docs/CoreAcademicSubjects.pdf seeded into a master_subjects table).
- Center Admin toggles which subjects their center offers, creating/removing center_subjects records.
- These subjects appear as options for teachers when creating courses.

Updated Navigation Rail

Nav Item	Available To
Dashboard	Both System Admin and Center Admin
Registrations (NEW)	System Admin only
Centers	System Admin (all) / Center Admin (own only)
Subjects (NEW)	Center Admin only
Users	Both (filtered by scope)
Content	Both (filtered by scope)
Analytics	Both (filtered by scope)

10. Updated Row Level Security (RLS) Policies

10.1 New Tables RLS

Apply the following RLS policies to all newly created tables:

Table	Student	Teacher	Center Admin	System Admin
courses	Read (enrolled only)	CRUD own	Read/Delete center's	All
course_enrollments	Read own / Insert own	Read (own courses)	Read center's	All
center_subjects	Read	Read	CRUD own center's	All
course_timeline	Read (enrolled)	CRUD (own courses)	Read	All
certificates	Read own	Read (own students)	Read center's	All
als_center_registrations	None	None	None	All (CRUD)

10.2 Update `current_user_role()` Function

Update the `current_user_role()` security definer function to handle all four roles: student, teacher, center_admin, system_admin. All existing policies that check `role = 'admin'` must be updated to check `role IN ('center_admin', 'system_admin')` or specifically `role = 'system_admin'` depending on the access level required.

11. Sync System Updates

11.1 New Entities to Include in Sync

The `SyncViewModel.syncAll()` method must be updated to include the following new entities in both push and pull operations:

Entity	Sync Direction
courses	Pull: students and teachers download courses they are associated with.
course_enrollments	Push: enrollment created offline syncs to cloud. Pull: fetch latest enrollment data.
center_subjects	Pull only: students/teachers download their center's subject list.
course_timeline	Pull: all timeline items for enrolled courses.
certificates	Pull: student's certificates downloaded for offline viewing.

11.2 SyncQueue Entry Types

Add the following operation types to the Drift `SyncQueue` `entityType` field:

- `course_enrollment` — when a student enrolls offline.
- `progress_update` — already exists, verify it is implemented.
- `course_create` — when a teacher creates a course offline.
- `timeline_item_create` — when a teacher adds a timeline item offline.

12. Resolving Known Issues & TODOs

The following items were listed as "Known Issues / TODOs" in the current README. Here are the resolution plans:

Issue #	Issue Description	Resolution Plan
#1	Biometric setup UI flow is partially implemented.	Complete the BiometricSetupView: after successful biometric enrollment, navigate to the dashboard. Ensure the biometric credential check on app startup calls biometricAutoFill() if isBiometricEnabled is true. Verify on both Android (fingerprint) and iOS (Face ID).
#2	Batch user import CSV validation could be more robust.	Add front-end validation before the API call: check required columns (email, full_name, role), validate email format, validate role values, check for duplicate emails within the batch file. Show a preview table before importing.
#3	Storage policies for profile pictures need refinement.	Set bucket policy on profile-pictures: Authenticated users can upload to their own userId path. Public read access for displaying profile photos. Deny overwrite from other users (path must start with auth.userId()).
#4 (New)	Two SQLite databases exist (sqlite and Drift), creating redundancy.	Long-term: migrate all sqlite tables to Drift for type-safety and better query building. Short-term: clearly document which database holds what. als_local.sqlite (Drift) = auth cache + SyncQueue. als_study_companion.db (sqlite) = all entity data. Do not mix them.
#5 (New)	admin role is ambiguous — it could mean Center Admin or System Admin.	Run a migration to split: rename existing "admin" users to "system_admin". Newly created Center Admin accounts use "center_admin". Update all CHECK constraints and RLS policies accordingly.

13. Implementation Roadmap

Phase 1 — Foundations (Do First)

These changes must be completed before any new features can be built. They set the groundwork.

51. Rename apps/student_phone → apps/mobile_app throughout the codebase (folder, pubspec.yaml, imports, CI scripts, README).
52. Rename packages/shared_models → packages/shared_core and packages/shared_services → packages/backend_services. Update all pubspec.yaml dependencies and import paths.
53. Update UserRole enum: add centerAdmin and systemAdmin values.
54. Run Supabase migration: alter users table CHECK constraint to include center_admin and system_admin.
55. Run Supabase migration: rename existing admin users to system_admin.
56. Update current_user_role() function to handle all four roles.
57. Update all RLS policies that reference role = 'admin' to use the correct new role names.

Phase 2 — Core New Tables & Models

58. Create Supabase migration for: als_center_registrations, courses, course_enrollments, center_subjects, course_timeline, certificates.
59. Add columns to existing tables: als_centers.registration_id, als_centers.center_admin_id, lessons.course_id.
60. Create all new Dart model files in shared_core/lib/models/.
61. Add new table name constants to DbConstants.
62. Add new SQLite table creation SQL to DatabaseHelper._createDb() for offline support.
63. Add new service methods to SupabaseDatabaseService and SupabaseStorageService in backend_services.

Phase 3 — ALS Center Registration (admin_web + public page)

64. Build public route /register-center in admin_web with the Center Registration Form.
65. Build CenterRegistrationReviewPage in admin_web for System Admin.
66. Implement approveCenter() and rejectCenter() logic including Center Admin account creation.
67. Update the ALS Center dropdown in mobile_app registration to pull from approved centers only.

Phase 4 — Course System (mobile_app)

68. Add `als_center_id` to Teacher and Student registration forms and include in `signUp user_metadata`.
69. Build `SubjectManagementPage` in `admin_web` for Center Admin.
70. Build `TeacherCourseCreateView` with subject selection, PIN generation, and QR code upload.
71. Build `TeacherCourseTimelineView` for adding timeline items.
72. Update `TeacherDashboardView` tabs (Lessons tab becomes Courses tab).
73. Build `StudentEnrollView` with QR scan and PIN entry.
74. Build `StudentCourseView` showing timeline, lessons, and live progress.
75. Implement course completion detection in `ProgressViewModel` and trigger certificate issuance.
76. Build `StudentCertificateView`.

Phase 5 — Polish & Fixes

77. Fix biometric setup UI flow (Issue #1).
78. Improve batch CSV import validation in `admin_web` (Issue #2).
79. Refine storage policies for `profile-pictures` bucket (Issue #3).
80. Update `SyncViewModel.syncAll()` to include all new entities.
81. Write integration tests for the enrollment flow (QR scan and PIN entry).
82. Write unit tests for `CourseModel`, `CourseEnrollmentModel`, `CertificateModel`.

14. Quick Reference Cheatsheet

14.1 Role → Platform → Dashboard

Role	Platform / Dashboard
system_admin	admin_web → SystemAdminDashboard. Full system access. Reviews center registrations.
center_admin	admin_web → CenterAdminDashboard. Scoped to their center. Manages subjects & approves teachers.
teacher	mobile_app → TeacherDashboardView. Creates courses, lessons, timelines. Views enrolled students.
student	mobile_app → StudentDashboardView. Enrolls via QR/PIN. Views lessons. Earns certificates.

14.2 Key File Locations After Rename

File / Path	Description
ALS-LMS/apps/mobile_app/lib/main.dart	Entry point for the mobile app (Students & Teachers).
ALS-LMS/apps/admin_web/lib/main.dart	Entry point for the web admin panel.
ALS-LMS/packages/shared_core/lib/	All models, enums, constants, utils.
ALS-LMS/packages/backend_services/lib/	All Supabase service wrappers and sync logic.
ALS-LMS/packages/shared_ui/lib/	Reusable widgets and design system.
supabase/migrations/	All SQL migration files, numbered sequentially.

14.3 Enrollment Flow Summary

Step	QR Code Path	PIN Path
1	Teacher opens Course Details in mobile_app.	Teacher shares the 6-char Course PIN verbally or via message.
2	Teacher shows QR code on screen or prints it.	Student opens mobile_app → Enroll → Enter PIN tab.
3	Student opens mobile_app → Enroll → Scan QR tab.	Student types the PIN and taps Confirm.
4	Student points camera at QR. App decodes the PIN automatically.	App validates PIN against courses table.
5	App calls enrollStudentInCourse(courseId, studentId).	App calls enrollStudentInCourse(courseId, studentId).

Step	QR Code Path	PIN Path
6	Student is navigated to StudentCourseView.	Student is navigated to StudentCourseView.

End of ALS Study Companion — Complete Structured Plan